

Image Resizing by Seam Carving and Image Matting

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Abstract—This project focuses on seamlessly inserting a foreground image into a background image using advanced image processing techniques. The methodology involves three critical steps: expanding the background image, extracting the foreground, and integrating the two. Initially, the background image is expanded using seam carving, a content-aware resizing technique that preserves essential visual elements by selectively removing or inserting pixels. Following this, the foreground is extracted from the original image using an image matting algorithm, which precisely separates the foreground from its background. The final step involves integrating the extracted foreground into the expanded background image, ensuring a natural and coherent insertion. The project demonstrates the effectiveness of these techniques in achieving high-quality image composites.

Index Terms—Image resizing, Seam carving, Image matting, Hole filling.

I. INTRODUCTION

In this project, we aim to seamlessly insert a foreground image into a background image using image processing techniques. The process involves three critical steps to ensure a natural and undistorted final result.

First, we expand the background image, referred to Figure 1, to create sufficient space between the cartoon characters. This expansion is achieved using the seam carving algorithm, which effectively adjusts the image size without distorting the characters or crucial elements within the image.

Next, we extract the foreground from Figure 2 using an image matting algorithm. Image matting allows us to precisely separate the foreground from the background, preserving intricate details such as fine edges.

Finally, we integrate the extracted foreground into the expanded background image. This step involves aligning and blending the foreground seamlessly with the background, ensuring that the insertion appears natural and coherent.



Figure 1: background



Figure 2: foreground

II. IMAGE RESIZING BY SEAM CARVING

Seam carving is a content-aware image resizing technique that allows for intelligent adjustment of an image's dimensions while preserving its essential visual content. Unlike traditional resizing methods, which uniformly stretch or shrink an image and often result in distortion, seam carving selectively removes or inserts pixels based on their importance to the overall image content ^[1].

Seam carving algorithm works in the following steps:

- Energy function calculation
- Seam identification
- Seam removal or insertion

A. Energy function calculation

To approaching content-aware resizing, we need to remove pixels in a judicious manner. Therefore, we need to find the unnoticeable pixels that blend with their surroundings.

In optional, we convert the background image into grayscale for color information is not essential in this process. We use sobel operator to calculate energy and detect the edges. This leads to the following simple energy function (1):

$$e(\mathbf{I}) = \left| \frac{\partial}{\partial x} \mathbf{I} \right| + \left| \frac{\partial}{\partial y} \mathbf{I} \right| \quad (1)$$

We can get the image like Figure 3. Optionally, we can utilize filter before sobel operator to reduce noise in background.

B. Seam identification

If we just remove or insert the pixels or the whole column with the lowest energy to resize the image, it may distort the important elements. Therefore, it is recommended to use seam carving algorithm to preserve the image content better.

Let $\mathbf{I}$ be an $n \times m$ image and a vertical seam can be defined as the following equation:

$$\mathbf{s}^x = \{s_i^x\}_{i=1}^n = \{x(i), i\}_{i=1}^n, s.t. \forall i, |x(i) - x(i-1)| \leq 1, \quad (2)$$



Figure 3: background processed with sobel operator

where x is a mapping $x : [1, \dots, n] \rightarrow [1, \dots, m]$. That is, a vertical seam is an 8-connected path of pixels in the image from top to bottom, containing one, and only one, pixel in each row of the image.

To find a optimal seam by dynamic programming, we can traverse the image from the second row to the last row and compute the cumulative minimum energy M for all possible connected seams for each entry (i, j) :

$$M(i, j) = e(i, j) + \min(M(i-1, j-1), M(i-1, j), M(i-1, j+1)) \quad (3)$$

After the process, the minimum value of the last row in M is the end of the optimal verticle seams. And we can backtrack to find the whole path of the optimal seam. We can get 20 minimum seams from the image, showing in the Figure 4, where the red lines are the optimal seams.

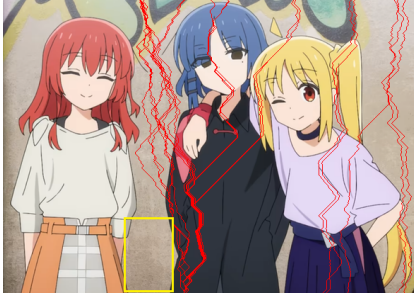


Figure 4: 20 optimal seams in the background

However, the result is not as consistent as expected. This inconsistency is due to excessive noise in certain areas, such as the highlighted in yellow frame in Figure 4, which may increase inessential energy in calculation. To address this problem, we can apply filter several times to reduce noise before sobel operation. We can get the result after applying 5 times filter, like Figure 5.

As we have already determined the location to be resized to insert foreground, we can apply seam carving only in certain area, like Figure 6.

C. Seam insertion

After determining the optimal seams, we can replicate the seams to expand the image, dedicating as Figure 7.



Figure 5: 20 optimal seams in the background with filter



Figure 6: 20 optimal seams in the background with filter and limitation

III. IMAGE MATTING

To separate the foreground from its original background, we use image hole filling based on morphological reconstruction and generate mask for foreground. The image hole filling algorithm involves the following steps:

- Initialize mask and marker images
- Morphological operation
- Iterative reconstruction process

A. Initialize mask and marker images

To get the mask image, we can binarize the original image to remove unimportant information unrelated to edge.

Marker image is an initial image indicating where the reconstruction should start, highlighting the area to be filled. In this case, for convenience, we can fill the background instead of the foreground, so that the marker image can start from the four corners of the image.

B. Morphological operation

The morphological operation used in image hole filling is dilation. Dilation is a fundamental morphological operation used in image processing, particularly in binary and grayscale image analysis. It is used to grow or expand the shapes in an image. The operation enhances the features of an image by adding pixels to the boundaries of objects.

For two sets $\mathbf{A}$ and $\mathbf{B}$, dilate $\mathbf{A}$ with $\mathbf{B}$ can be defined as:

$$\mathbf{A} \oplus \mathbf{B} = \{z | (\hat{\mathbf{B}})_z \cap \mathbf{A} \neq \emptyset\} \quad (4)$$

The foreground element of $\mathbf{B}$ overlaps with at least one element



Figure 7: the expanded background

of $\mathbf{A}$, and $\mathbf{z}$ is the shift of the origin of $\mathbf{B}$.

C. Iterative reconstruction process

We need to perform morphological reconstruction by iteratively propagating values from the marker image into the holes of the mask image. This process continues until stability is reached, meaning no further changes occur between iterations. This process can be indicated as Function 5:

$$\mathbf{X}_k = (\mathbf{X}_{k-1} \oplus \mathbf{B}) \cap \mathbf{I}^C, k = 1, 2, 3, \dots \quad (5)$$

$\mathbf{I}$ is the image to be processed, $\mathbf{X}_k$ is the image after k iterations, $\mathbf{B}$ is a symmetric structure element for 4-connection.

At the end of this process, we can minus the processed image by the original image to separate foreground from background. We can access the mask like Figure 8.



Figure 8: mask for foreground

Applying the mask to the image, we can generate a result like Figure 9. It is not ideal for residual background around the edge, because of error in edge detection. To address this problem, we can apply dilation to generate an ideal image like Figure 10.

IV. FINAL RESULT

Finally, we can scale the background and foreground, and merge them into one image, as Figure 11.



Figure 9: foreground



Figure 10: optimized foreground



Figure 11: final result

REFERENCES

- [1] S. Avidan and A. Shamir, "Seam carving for content-aware image resizing," *ACM Trans. Graph.*, vol. 26, p. 10-es, jul 2007.